CHI SQUARE for Goodness of Fit

In 2010 census of the city, the weight of the individual in a small city were found to be the following.

<50kg	50-75	>75
20%	30%	50%

In 2020, weight of $n = 500$ individuals were sampled. Below are the results.

<50	50-75	>75
140	160	200

Using $\alpha = 0.05$, would you conclude the population difference of weights changed in the last 10 years?

Ans. 2010 Expected

<50kg	50-75	>75
20%	30%	50%

2020 $n=500$ Observed

<50	50-75	>75
140	160	200

<50	50-75	>75
$0.2 \times 500 = 100$	0.3×500 150	0.5×500 250

I. Null hypothesis (H_0): The data meets the expectations.
Alternate hypothesis (H_1): The data does not meet the expectations.

II. $\alpha = 0.05$ CI = 95%.

III. Degree of freedom

$$df = K - 1 = 3 - 1 = 2$$

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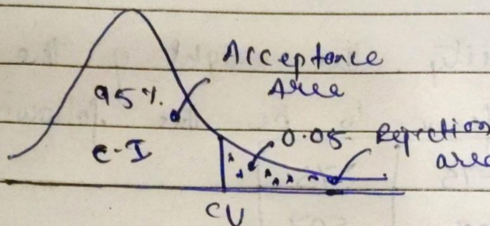
iv. Decision Boundary



CV – Critical value

CV = 0.5991

If X^2 is greater than 5.991, Reject Ho else we fail to reject the Null hypothesis

v. Calculate the Chi-Square Test Statistics

$$X^2 = \Sigma \frac{(O-E)^2}{E}$$

$$= \frac{(140-100)^2}{100} + \frac{(160-150)^2}{100} + \frac{(200-250)^2}{100}$$

$$= 16 + 0.66 + 10$$

$$X^2 = 26.66$$

$X^2 = 26.66 > 5.99 \Rightarrow$ Reject Ho.

Answer:

The weight of 2020 population are different than those expected in the 2010 population.